

ZF + GCH Set Theory

A collection of notes on Zermelo-Frankel set theory,
axiomatic definition of ordinals, cardinals, and their
relation with the generalized continuum hypothesis, the
axiom of choice and with math in general.

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1 Functions

Math itself relies heavily on the concept of function. In this section we will define a function in an extremely formal way, using the fundamental axioms of set theory to back-up our definitions and theorems right from the start.

1.1 Formal construction of a function

Building a function from the ground up is a bit tricky, since we need to define a function in terms of sets. We will first define a tuple, then a relation and finally a function. We will also define the concepts of injectivity and surjectivity.

Definition 1 (Tuple). *We can define a tuple like $(a_0, a_1, a_2, \dots, a_n)$ as a set, since everything in set-theory is just sets. In particular, we can make the following definition:*

$$(a_0, a_1, a_2, \dots, a_n) = \{\{0, \{a_1\}\}, \dots, \{n, \{a_n\}\}\}$$

Definition 2 (Relation). *A relation is basically a boolean predicate with two arguments. We can define a relation R as the set of tuples that satisfy the relation. For example, the relation R can be defined as a subset of the cartesian product of two sets A and B (the domain and codomain of the relation):*

$$a R b \iff (a, b) \in R \subseteq A \times B$$

Definition 3 (Function). *A function is a special case of a relation, where every element in the domain has exactly one image in the codomain.*

$$\begin{aligned} a f b &\iff b = f(a) \\ f : A \rightarrow B &= \{(x, f(x)) \in A \times B\} \end{aligned}$$

Definition 4 (Injection). *An injective function, also called injection, is a function that maps distinct elements of the domain to distinct elements of the codomain.*

$$\begin{aligned} \forall x_1, x_2 \in A : f(x_1) = f(x_2) &\Rightarrow x_1 = x_2 \\ \nexists x_1 \neq x_2 \in A : f(x_1) = f(x_2) \end{aligned}$$

Definition 5 (Surjection). *A surjective function, also called surjection, is a function such that no elements of the codomain are unreachable.*

$$\forall y \in B \exists x \in A : f(x) = y$$

Definition 6 (Bijection). *A bijective function, also called bijection, is a function that is both an injection and a surjection. Note that a bijection is a function that maps distinct elements of the domain to distinct elements of the codomain and every element in the codomain is reachable by some element in the domain.*

Definition 7 (Image). We call $\vec{f}(S)$ the image of a given set S under the function f . The image of a set is the set of the elements in the codomain mapped by elements in S via the function f .

$$\vec{f}(S) = \{f(x) \in B : x \in S\}$$

Definition 8 (Anti-image). We call $\overleftarrow{f}(T)$ the anti-image of a given set T under the function f . The anti-image of a set is the set of the elements in the domain mapped to elements in T via the function f .

$$\overleftarrow{f}(T) = \{x \in A : f(x) \in T\}$$

Lemma 1. The anti-image of a singleton set is either a singleton set or the empty set for every injection.

Proof. Let $A = \overleftarrow{f}(\{y\})$ be the set of all the $x \in A$ such that $f(x) = y$. We can apply the definition of injection and say that if $f(x_1) = f(x_2)$ then $x_1 = x_2$. Therefore if $x_1, x_2 \in A \Rightarrow x_1 = x_2$, so either $A = \{x\}$ or $A = \emptyset$. □

Lemma 2. The anti-image of a singleton set is never empty for every surjection.

Proof. Let $A = \overleftarrow{f}(\{y\})$ be the set of all the $x \in A$ such that $f(x) = y$. We can apply the definition of surjection and say that an x such that $f(x) = y$ can always be found. Hence such x is in A which therefore can't be empty. □

Lemma 3. The image of a singleton set is a singleton for every bijection f . Hence a function f^{-1} can be defined as $f^{-1}(y) = \zeta_1(\overleftarrow{f}(\{y\}))$ where $\zeta_1(\{x\}) = x$.

Notice that, $f^{-1} \circ f = id_A$ and $f \circ f^{-1} = id_B$, we call this function the inverse of f .

Lemma 4. The images of disjoint sets are disjoint under any injection.

Proof. Let A_1 and A_2 be two disjoint sets. We can apply the definition of injection and say that if $f(x_1) = f(x_2)$ then $x_1 = x_2$. Therefore if $x_1 \in A_1$ and $x_2 \in A_2$ then $f(x_1) \neq f(x_2)$, this implies that $\vec{f}(A_1)$ and $\vec{f}(A_2)$ are disjoint since the generic element of $\vec{f}(A_1)$ is in the form $f(x_1)$ and the generic element of $\vec{f}(A_2)$ are in the form $f(x_2)$ and such elements are never equals under the assumptions that x_1 and x_2 are elements of disjoint sets hence they're not equal. □

1.2 Functions and power-sets

Power-sets are a very important concept in set theory. The set $P(A)$ is the set of all subsets of a given set A , and it's called power-set. The existence of a power-set for a given set A is guaranteed by the axiom of power-set. This is done because we can't express the power-set as $P(A) = \{X \in \mathbb{U} : X \subseteq A\}$, because we can't find an upper bound $\mathbb{U}$ for every power-set, hence we would violate axiom of schema-specification.

Theorem 1. *Given a set A , the power-set $P(A)$ is in bijection with the set of all the functions $c : A \rightarrow \{0, 1\}$. The function c associated with a given subset S is called the characteristic function of the subset S .*

Proof. We can exhibit a bijection f between every possible S subset of A and every possible function $c : A \rightarrow \{0, 1\}$ constructed as follows:

$$f(S) = c(x) = \begin{cases} 1 & \text{if } x \in S \\ 0 & \text{otherwise} \end{cases}$$

We can show that is a bijection by also exhibiting the inverse function f^{-1} constructed as:

$$\begin{aligned} f^{-1}(c) &= \{x \in A : c(x) = 1\} \\ f^{-1}(c) &= \overleftarrow{c}(\{1\}) \end{aligned}$$

□

Please, notice that we must be able to construct the set of all functions in the form $f : A \rightarrow B$ to be able to make sense out of the previous theorem. We can do it respecting the axiom of schema-specification by using the formal definition of function as a set of tuples for which every element in the domain only appear once.

$$\mathbb{F}_{A \rightarrow B} = \{f \in P(A \times B) : (a, y), (b, y) \in f \Rightarrow a = b\}$$

Theorem 2 (Cantor's theorem). *There is no suriection from a set to its power-set.*

Proof. We can prove this by contradiction. Let's assume that there is a suriection $f : A \rightarrow P(A)$. We can construct the set $S = \{x \in A : x \notin f(x)\}$. By the definition of suriection, we know that there is an element $y \in A$ such that $f(y) = S$. Now we can ask ourselves if y is in S or not:

$$y \in S \iff y \in f(y) \iff y \notin S$$

Hence, we conclude that there is no suriection from a set to its power-set, since that what originated the contradiction. By assuming that there is no y such that $f(y)$ is equal to the set S , we can escape this apparent paradox hence we conclude that no function from a set to its power-set can ever be a suriection.

□

2 Additional sets constructions

In this chapter we will introduce some additional constructions of sets that are important in set theory, namely the Von Neumann rank and the Hartogs' ordinal.

2.1 Von Neumann rank

We can construct an hierarchy of sets, well ordered with respect with $\subseteq$, called the Von Neumann hierarchy, in such a way that any well-formed set in ZF is subset of at least one of the sets in this hierarchy. We can define the Von Neumann rank of a set A in terms of the Von Neumann hierarchy as follows:

Definition 9 (Von Neumann universe). *For every ordinal i we define the Von Neumann universe V_i as the set which obey the following construction:*

$$V_i = \begin{cases} \emptyset & \text{if } i = 0 \\ \bigcup_{j < i} P(V_j) & \text{otherwise} \end{cases}$$

Theorem 3 (Von Neumann theorem). *Every set S is subset of some universe V_i*

Proof. Every entity in set theory is a set, moreover. Every set is either empty or is a set containing other sets. Let's proceed by induction. The base case is V_0 which is the empty set. Assume all the element of some set S are subsets of some Von Neumann universe V_j , we can assert that S is subset of V_{j+1} by the definition of V_{j+1} . $\square$

Definition 10 (Von Neumann rank). *For every set S we can define $r(S)$ the Von Neumann rank of S as the smallest ordinal i such that $S \subseteq V_i$.*

2.2 Hartogs' ordinal

The Hartogs' ordinal is a special ordinal that can be used to prove the existence of a well ordering for any set under certain assumptions. It can be defined using axioms of ZF alone. Specifically, its existence does not depend on the Axiom of Choice (AC).

Definition 11 (Hartogs' ordinal). *The Hartogs' ordinal α of a set A is the smallest ordinal α such that there exists an injection from A to α . In other words, it is the smallest ordinal that is "larger" than A .*

Theorem 4 (Hartogs' theorem). *For every set A there exists an Hartogs' ordinal.*

Proof. If A is empty, then the Hartogs' ordinal is 0. Otherwise, we can construct the Hartogs' ordinal by taking the set of all ordinals that are injectable into A . The set that is constructed this way is itself an ordinal, and since it doesn't belong to the set itself it must not be injectable into A . Therefore, it is the Hartogs' ordinal of A . $\square$

3 Cardinality

The most difficult part of set theory is the concept of cardinality. The cardinality of a set is a measure of the "size" of the set, and it can be defined in several different ways. In this section we will define the concept of cardinality in a formal way, providing two definitions, one that works only if AC (Axiom of Choice) is assumed to be true and one that works even if AC is neither assumed true nor false.

3.1 Fundamentals of cardinality

Regardless of the definition of cardinality that we choose to use, we denote $|A|$ the cardinality of A . We will propose two different definition for cardinality. We will use the notation $|A| \rightleftharpoons |B|$ to denote that $|A|$ and $|B|$ are in mutual injection, thus that both an injection from A to B and an injection from B to A exist. Bijection clearly implies mutual injection. The Bernstein's theorem (theorem 7) states that if there is a mutual injection implies mutual bijection, but right now, we can't prove it yet.

Definition 12 (Cardinality with AC). *The cardinality of a set A is the smallest ordinal in bijection with A , Assuming well ordering is essential to this definition.*

Definition 13 (Scott's cardinality). *The cardinality of a set A is the set of all sets in mutal injection with A that also have minimal rank.*

Definition 14 (Cardinality comparison). *We can compare cardinalities using $\leq$:*

$$|A| \leq |B| \iff \exists f : A \rightarrow B \text{ injective}$$

Definition 15 (Cardinal sum). *We can define the sum for cardinal arithmetic as the following operation, which is defined as the cardinality of the "disjoint union":*

$$|A| + |B| = |(\{0\} \times A) \cup (\{1\} \times B)|$$

Definition 16 (Cardinal product). *We can define the sum for cardinal arithmetic as the following operation, which is defined as the cardinality of the set product:*

$$|A| \times |B| = |A \times B|$$

Definition 17 (Cardinal exponentiation). *We can define the sum for cardinal arithmetic as the following operation, which is defined as the cardinality of the set of all functions having domain and codomain as the operands:*

$$|A|^{|B|} = |f : B \rightarrow A|$$

Please be very aware that using the definition of cardinality of a given set as the smallest ordinal in bijection with such set, you are still supposed to use the cardinal arithmetic operations defined above. Ordinal arithmetic is not the same as cardinal arithmetic, and, since not every ordinal is a cardinal, we can't use the ordinals arithmetic operations on cardinals. For example $\omega \neq \omega + 1$ but $|A| + 1 = |A|$ for every set A in bijection with omega.

3.2 Scott's cardinality in depth

Scott's cardinality is a way to define the cardinality of a set that is independent from AC. We made use of the concept of rank in the definition of cardinality. Specifically, the cardinality of a set A is the set of all sets in mutual injection with A that also have minimal rank. Why is that? because if we were to consider the set of all sets in mutual injection with A regardless of their rank, we would end up with an ill-formed set.

Lemma 5. (*Wrapping lemma*) *Let A be a well-ordered infinite set, then, then $A \cup \{X\}$ is in bijection with A for every possible choice of a generic set X .*

Proof. Let α be the ordinal in bijection with $A : \alpha \rightarrow (\alpha \cup \{X\})$ as follows:

$$f(x) = \begin{cases} x' & \text{if } x = x' + 1 \\ \{X\} & \text{if } x = 0 \end{cases}$$

We can exhibit an inverse for this function herby proving its bijectivity by construction:

$$f^{-1}(x) = \begin{cases} x + 1 & \text{if } x \in \alpha \\ 0 & \text{if } x = \{X\} \end{cases}$$

Since A is an infinite well-ordered set, it is in bijection with some α hence some $(\alpha \cup \{X\})$ is in bijection with A as well. This result can be thought of as $|A| + 1 = |A|$ for every well ordered infinite set. □

Lemma 6. (*Ill-formedness of rankless definition of cardinality*) *The class of every set in mutual injection with a given set is not a set for every well-ordered infinite set.*

Proof. Let A be a well-ordered infinite set, let A^* be the class of all sets in mutual injection with A . Notice that $A \cup \{X\}$ is in bijection with A for every possible choice of a generic set X . Hence, it is in A^* . Now, consider the following construction:

$$A \cup \{X\} \in A^*$$

$$\{X\} \subseteq \bigcup_{S \in A^*} S$$

$$X \in P\left(\bigcup_{S \in A^*} S\right)$$

This means that assuming that A^* is a set, led us to a way of creating the universal set, which itself gives us a way to fall within the russel paradox. This means that A^* is not a well-formed set, hence we can conclude that the class of all sets in mutual injection is not a set for every well-ordered infinite set. Under AC, this generalizes to the fact that no class of sets in bijection with any infinite sets can ever be a set. □

One could ask himself why is that the case that the previous argument doesn't apply to the proper Scott's cardinality definition that takes the rank into account. The reason is simple: assuming that S is a set of minimal rank, it's not the case that $S \cup \{X\}$ is also of minimal rank. We can exhibit a counterexample for this:

$$\begin{aligned} V_\omega &\subseteq V_\omega \\ V_\omega \cup \{V_\omega\} &\not\subseteq V_\omega \end{aligned}$$

We took a set of rank ω of minimal rank V_ω and we constructed another set, $V_\omega \cup \{V_\omega\}$, which is not of minimal rank while still being in bijection with V_ω . This means that the previous argument will fail under the rank-aware definition of Scott's cardinality.

Lemma 7. (*Transitivity of cardinality comparison*) $|A| \leq |B| \wedge |B| \leq |C| \Rightarrow |A| \leq |C|$.

Proof. Let f_{AB} be the injection from A to B and f_{BC} be the injection from B to C . We can construct $f_{AC} = f_{BC} \circ f_{AB}$, which is an injection from A to C , thus $|A| \leq |C|$. $\square$

Lemma 8. (*Transitivity of mutual injection*) $|A| \rightleftharpoons |B| \wedge |B| \rightleftharpoons |C| \Rightarrow |A| \rightleftharpoons |C|$.

Proof.

$$\begin{aligned} |A| \rightleftharpoons |B| \rightleftharpoons |C| &\Rightarrow |A| \leq |B| \leq |C| \\ |C| \rightleftharpoons |B| \rightleftharpoons |A| &\Rightarrow |C| \leq |B| \leq |A| \\ \therefore |A| &\rightleftharpoons |C| \end{aligned}$$

$\square$

Lemma 9. (*Correctness of Scott's cardinality*) *Even considering only sets of minimal rank, two sets in mutual injection still have the same cardinality regardless of their ranks.*

Proof. For every two sets A and B , them being in mutual injection implies that they have the same cardinality. In fact, Let $|A|$ be the set of all sets in mutual injection with A that also have minimal rank. May $|B|$ also be the set of all sets in mutual injection with B that also have minimal rank. every set in $|A|$ is in mutual injection with A , which is in mutual injection with B , which is in mutual injection with every set in $|B|$.

$$\forall X \in |A| \quad X \rightleftharpoons A \wedge A \rightleftharpoons B \wedge B \rightleftharpoons Y \quad \forall Y \in |B|$$

$\square$

Definition 18 (Successor cardinality). *The successor cardinality k^+ of a given cardinality k is the smallest possible cardinality strictly greater than k under cardinality-comparison.*

Lemma 10 (Existence of the successor cardinality). *Every cardinality has a successor.*

Proof. The Cantor's theorem (theorem 2) states that for every set A , the power set of A is strictly greater than A itself. Hence, no cardinality can ever be the greatest one. $\square$

3.3 Generalized Continuum Hypothesis (GCH)

An additional axiom that can be added to ZF is the Generalized Continuum Hypothesis (GCH). It states that for every cardinality κ , its successor is 2^κ . This is what we will assume from now on. We will show how this axiom can be used to prove that every set can be well ordered (e.g. it can be put in bijection with some ordinal).

3.4 Sierpinski's theorem

The Sierpinski's theorem states that the GCH implies that for every infinite set A the cardinality of the cartesian product of A with itself is the same as the cardinality of A itself. This means that $A \times A \sim A$ for every infinite set A .

Lemma 11. *Let k be a cardinality of an infinite set, then $k+1 = k$ under the assumption that the generalized continuum hypothesis holds true at least for k (local GCH for k).*

Proof.

$$\begin{aligned} k &\leq k+1 < 2^k \\ \text{GCH}(k) : \exists \bar{k} : k &\leq \bar{k} < 2^k \Rightarrow \bar{k} = k \\ \therefore k+1 &= k \end{aligned}$$

□

Lemma 12. *Let k be a cardinality of an infinite set, then $2k = k$ under the assumption that the generalized continuum hypothesis holds true at least for k (local GCH for k).*

Proof.

$$\begin{aligned} k &\leq k+k < 2^k + 2^k = 2^{k+1} = 2^k \\ \text{GCH}(k) : \exists \bar{k} : k &\leq \bar{k} < 2^k \Rightarrow \bar{k} = k \\ \therefore 2k &= k \end{aligned}$$

□

Theorem 5 (Sierpinski's theorem). *Let k be a cardinality of an infinite set, then $k^2 = k$ under the assumption that the generalized continuum hypothesis holds true at least for k .*

Proof.

$$\begin{aligned} k &\leq k^2 < (2^k)^2 = 2^{2k} = 2^k \\ \text{GCH}(k) : \exists \bar{k} : k &\leq \bar{k} < 2^k \Rightarrow \bar{k} = k \\ \therefore k^2 &= k \end{aligned}$$

□

Notice how we can assume that GCH holds true in general and then prove that for every infinite set A with cardinality k , since $k = k^2$ then $A \times A \sim A$. This clearly holds true for $\mathbb{N}$, but with the GCH this also applies to $\mathbb{R}$ and to every infinite set in general.

3.5 Tarski's well ordering theorem

Assuming that for every set A , assuming it's infinite, then $A \times A \sim A$, we can prove that the axiom of choice holds true by first proving that every set can be well ordered. We will hence discuss the Tarski's proof of the well ordering theorem.

Theorem 6 (Tarski's well ordering theorem). *Assuming that every infinite set A has the property that $A \times A \sim A$, then every set can be well ordered.*

Proof. Of course every finite set can be well ordered, since a bijective function between such set and a proper finite subset of the natural numbers can be created, and since every finite subset of natural numbers is also well ordered. Hence, without loss of generality, we can assume that the set A is infinite.

Let A be an infinite set and let β be the Hartogs ordinal for A . Which means that β is the smallest ordinal that cannot be injected into A (So the smallest ordinal that is greater than $|A|$). Such ordinal can be constructed as the transitive set of all ordinals with cardinality greater than or equal to $|A|$ as shown in the proof of the Hartogs theorem.

Let $A' := A \cup \beta$ be the union of the infinite set A and its Hartogs ordinal. Since A is infinite, then A' is also infinite, and since A' is infinite, under our assumption, $A' \times A' \sim A'$.

For every $a \in A$ we can construct the family S_a which has some interesting properties.

$$S_a := \vec{f}(\{a\} \times \beta) \cap \beta$$

Since f is a bijection, then the images of two disjoint sets are also disjoint. This means that the family S_a is a family of disjoint sets. Since $|(\{a\} \times \beta)| = |\beta| > |A|$ then its image through f is also greater than $|A|$ given that f is a bijection.

$$\begin{aligned} |\vec{f}(\{a\} \times \beta)| &> |A| \\ \vec{f}(\{a\} \times \beta) &\subseteq A \cup \beta \\ \therefore \vec{f}(\{a\} \times \beta) \cap \beta &\neq \emptyset \end{aligned}$$

We can finally define the relation $a \leq b \iff \min(S_a) \leq \min(S_b)$ which is a total ordering over the set A . We can then assert that since the set A is well-ordered this means that A is in bijection with some ordinal.

□

4 Consequences of the Axiom of Choice

Whether we assume the existence of a choice function or we prove its existence under the assumption that the generalized continuum hypothesis is true, we can then prove some really interesting theorems and lemmas that turns out to be very useful not only in set theory, but in math as a whole.

4.1 Formal notion of invertibility

In mathematics we often end up talking about invertible functions. In this section we will define the concept of invertibility and bijectivity and we will prove that a function is invertible if and only if it is bijective.

Definition 19 (Left inverse). *A function $g : B \rightarrow A$ is the left inverse of $f : A \rightarrow B$ if and only if $g \circ f = \text{id}_A$, where id_A is the identity function on the set A .*

$$\forall x \in A : g(f(x)) = x$$

Definition 20 (Right inverse). *A function $h : B \rightarrow A$ is the right inverse of $f : A \rightarrow B$ if and only if $f \circ h = \text{id}_B$, where id_B is the identity function on the set B .*

$$\forall y \in B : f(h(y)) = y$$

Lemma 13. (Uniqueness of the inverse) *If a function $f : A \rightarrow B$ has a left inverse $g : B \rightarrow A$ and a right inverse $h : B \rightarrow A$, then $g = h$.*

$$g = g \circ \text{id}_A = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_B \circ h = h$$

Lemma 14. (Existence of the left inverse implies injectivity) *If a function $f : A \rightarrow B$ has a left inverse $g : B \rightarrow A$ that function is injective.*

$$\exists g : g \circ f = \text{id}_A \Rightarrow [f(x_1) = f(x_2) \Rightarrow g(f(x_1)) = g(f(x_2)) \Rightarrow x_1 = x_2]$$

Lemma 15. (Existence of the right inverse implies surjectivity) *If a function $f : A \rightarrow B$ has a left right $h : B \rightarrow A$ that function is surjective.*

$$\exists h : f \circ h = \text{id}_B \Rightarrow [\forall y \in B \exists x = h(y) : f(x) = y]$$

Lemma 16. (Injectivity implies existence of the left inverse) *Assuming the existence of a function ζ that returns a single element of a given set, called the choice function, we can prove that if a function $f : A \rightarrow B$ is injective, then it has a left inverse.*

$$g(y) = \begin{cases} x \in A : f(x) = y & \text{if it exists} \\ \zeta(A) & \text{otherwise} \end{cases}$$

Lemma 17. (Injectivity implies existence of the left inverse) *Assuming the existence of a function ζ that returns a single element of a given set, called the choice function, we can prove that if a function $f : A \rightarrow B$ is surjective, then it has a right inverse.*

$$h(y) = \zeta(\overleftarrow{f}(\{y\}))$$

4.2 Bernstein's theorem

One of the most impressive results in set theory is the so-called Bernstein's theorem. The Bernstein theorem basically states that even if we might not be able to exhibit a bijection between two sets, such bijection must exist under certain conditions. This will later be useful to study the so-called "cardinality" of a set.

Theorem 7 (Bernstein's theorem). *Given two sets A and B , if there is an injection from $f : A \rightarrow B$ and an injection from $g : B \rightarrow A$, then there is a bijection between the two.*

Proof. First, define the following entities (needed for the construction of the bijection):

$$\begin{aligned} A_0 &= A - \vec{g}(B) \\ A_{n+1} &= \vec{g}(\vec{f}(A_n)) \\ A_\infty &= \bigcup A_i \end{aligned}$$

First of all, we need to understand that $g : B \rightarrow A$ is an injection, thus the very same function can be treated as a bijection when the codomain is restricted to the image of the domain. We can write it in symbols with the following formula:

$$g : B \rightarrow A \quad \text{injection} \Rightarrow g : B \rightarrow \vec{g}(B) \quad \text{bijection}$$

Since g is a bijection, we can claim that $g^{-1} : \vec{g}(B) \rightarrow B$ exists and is a bijection. Thus, once again, we can write the statement in symbols as the following formula:

$$g : B \rightarrow A \quad \text{injection} \Rightarrow g^{-1} : \vec{g}(B) \rightarrow B \quad \text{bijection}$$

We can rewrite it substituting $\vec{g}(B)$ with $A - A_0$, since that's the very definition of A_0 :

$$g : B \rightarrow A \quad \text{injection} \Rightarrow g^{-1} : (A - A_0) \rightarrow B \quad \text{bijection}$$

We can finally restrict the domain of g^{-1} to the set $A - A_\infty$, which is a subset of $A - A_0$, hence we can write the following statement:

$$g : B \rightarrow A \quad \text{injection} \Rightarrow g^{-1} : (A - A_\infty) \rightarrow B \quad \text{injection}$$

We can construct the bijection $h : A \rightarrow B$ and prove that it is indeed a bijection (we might not be able to use it to actually compute the value of $h(x)$ since, in order to do it, we might need infinite computation power, but we can prove that at least it exists):

$$h(x) = \begin{cases} f(x) & \text{if } x \in A_\infty \\ g^{-1}(x) & \text{otherwise} \end{cases}$$

We can also observe that $g : B \rightarrow A - A_\infty$ is also a bijection since $A - A_\infty$ is a subset of $A - A_0$, hence h is also a bijection for all $x \in A - A_\infty$ due to its very construction.

We can prove that h is an injection by proving that $h(x_1) = h(x_2)$ implies $x_1 = x_2$. We have to reason about two cases, both of which leads to injectivity.

$$\begin{aligned} \forall x_1, x_2 \in A_\infty [h(x_1) = f(x_1) = f(x_2) = h(x_2) \Rightarrow x_1 = x_2] \\ \forall x_1, x_2 \notin A_\infty [h(x_1) = g^{-1}(x_1) = g^{-1}(x_2) = h(x_2) \Rightarrow x_1 = x_2] \end{aligned}$$

Please notice that a scenario where x_1 and x_2 are not either both in A_∞ or both are not is actually impossible under the assumption that $h(x_1) = h(x_2)$, because:

$$\begin{aligned} \forall x_1 \in A_\infty, \exists x_2 \in A \Rightarrow [f(x_1) = h(x_1) = h(x_2) = g^{-1}(x_2)] \\ \therefore f(x_1) = g^{-1}(x_2) \Rightarrow g(f(x_1)) = x_2 \Rightarrow x_2 \in A_\infty \end{aligned}$$

We can also prove that h is a surjection by proving that for every $y \in B$ there is an $x \in A$ such that $h(x) = y$. To find such x , we can reason about the following two cases:

$$\begin{aligned} g(y) \notin A_\infty \Rightarrow h(g(y)) = g^{-1}(g(y)) = y \\ g(y) \in A_\infty \Rightarrow g(y) \in \vec{g}(\vec{f}(A_n)) \Rightarrow f(\zeta(\vec{f}(A_n))) = y \end{aligned}$$

It's worth noting that $g(y) \in A_\infty$ implies that $g(y) \in A_\infty - A_0$ since A_0 is literally defined as $A - \vec{g}(B)$. Hence, we can assume $g(y) \in A_{n+1}$

By the definition of $A_{n+1} = \vec{g}(\vec{f}(A_n))$, we can say that $x = g(y) \in A_{n+1}$ implies that $y = \vec{f}(A_n)$, which means we can claim that the following statement holds true:

Please notice that we are once again assuming a choice function ζ that returns a single element of a given set exist, while it's easy to prove that it exist case by case, its existence cannot be proved in general. This means that the Bernstein's theorem is not provable in ZF set theory alone, but it is provable in ZFC set theory (ZFC + AC).

□